



microring modulators.

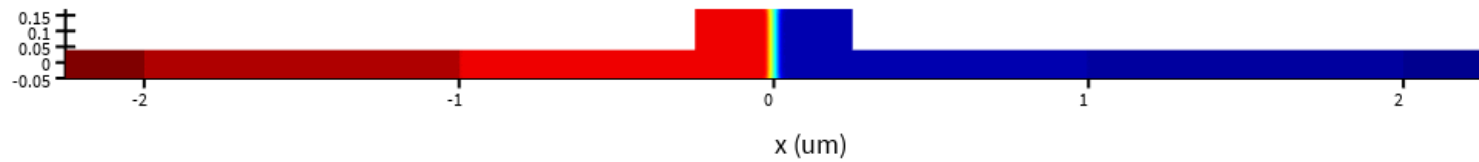


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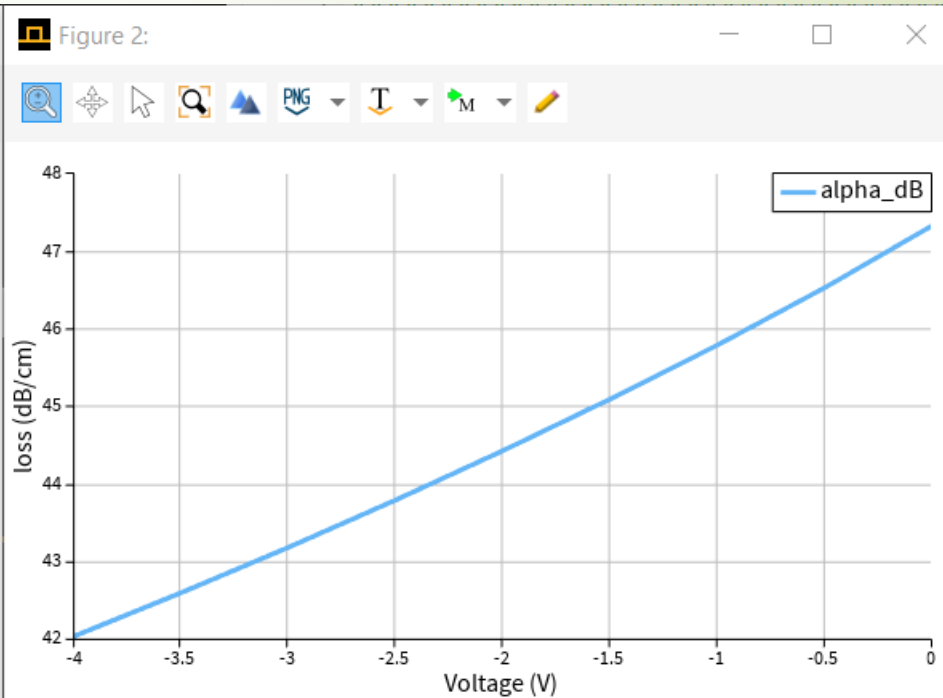
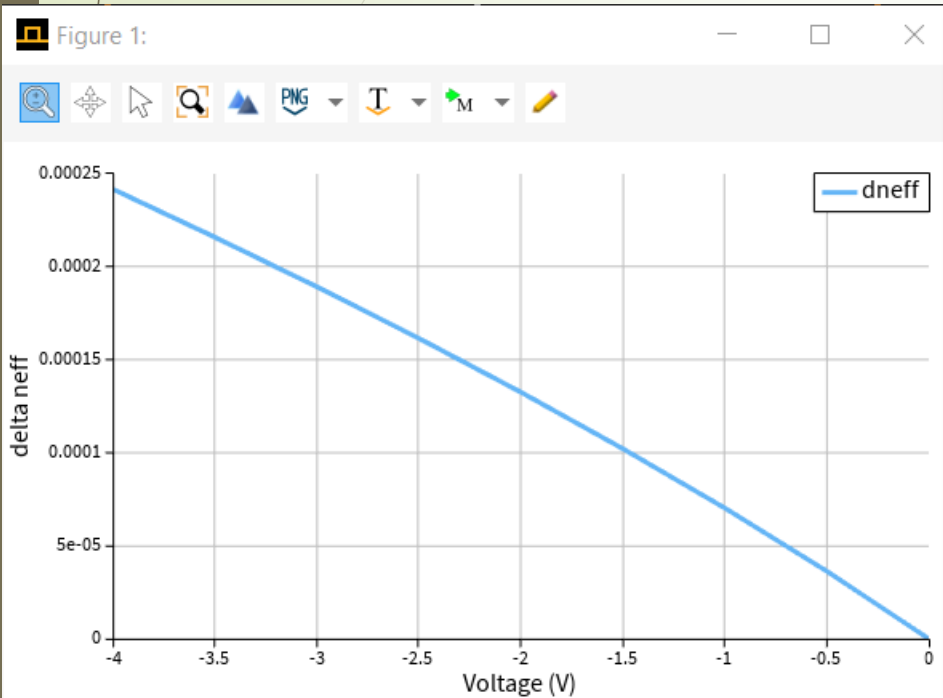
Ring Resonator Simulation

- ➡ Lateral PN junction.

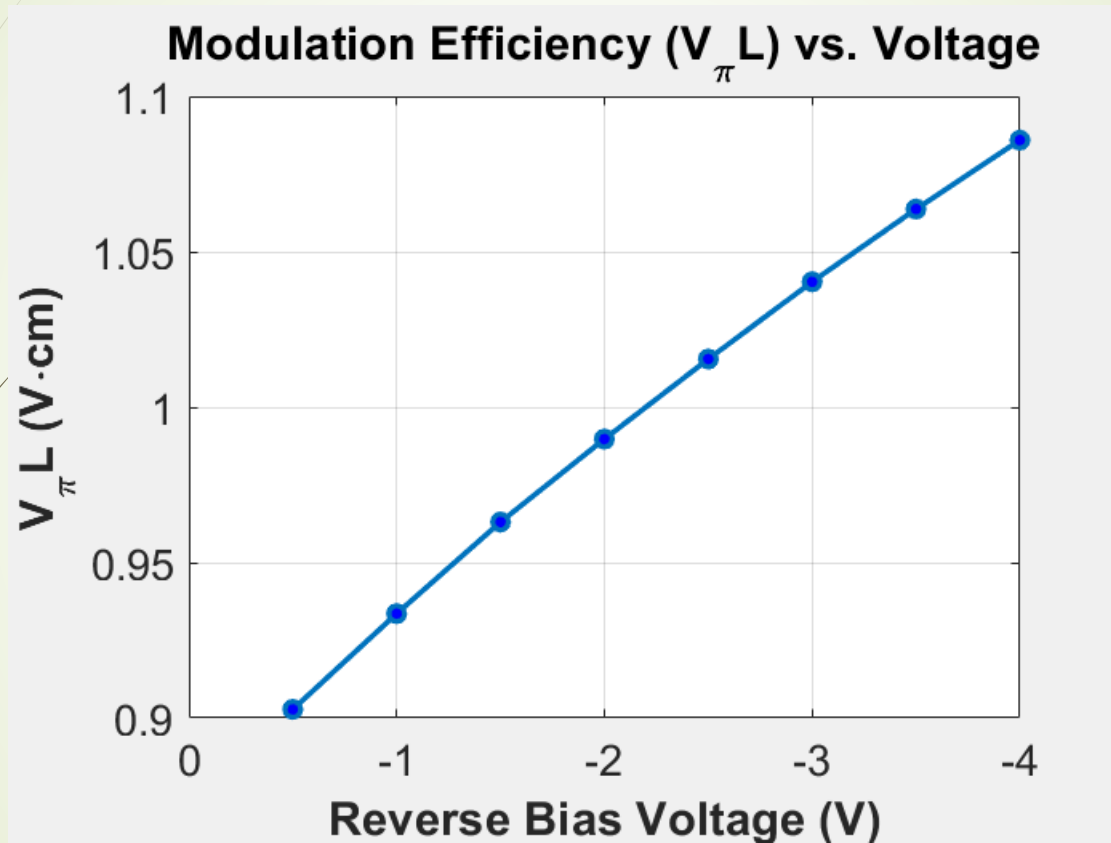
Lateral PN junction Simulation



Lateral PN junction Simulation



➤ Lateral PN junction Simulation





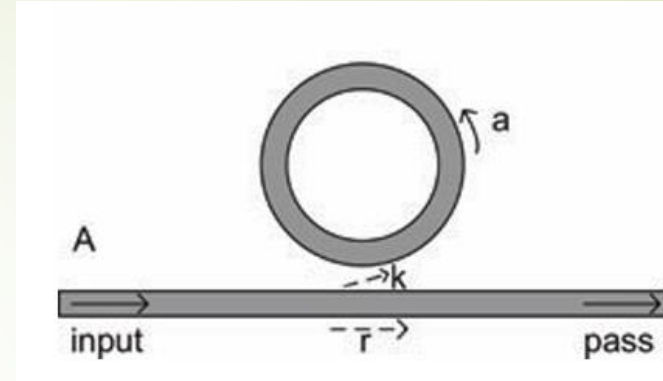


➤ Lateral PN junction used for modulation

- **FSR = 7.234621 nm,**
- **FWHM = 0.141260 nm,**
- **Q = 9258,**
- **efficiency = 54 pm/V**

- Lateral PN junction used for modulation

$$r^2 + k^2 = 1$$



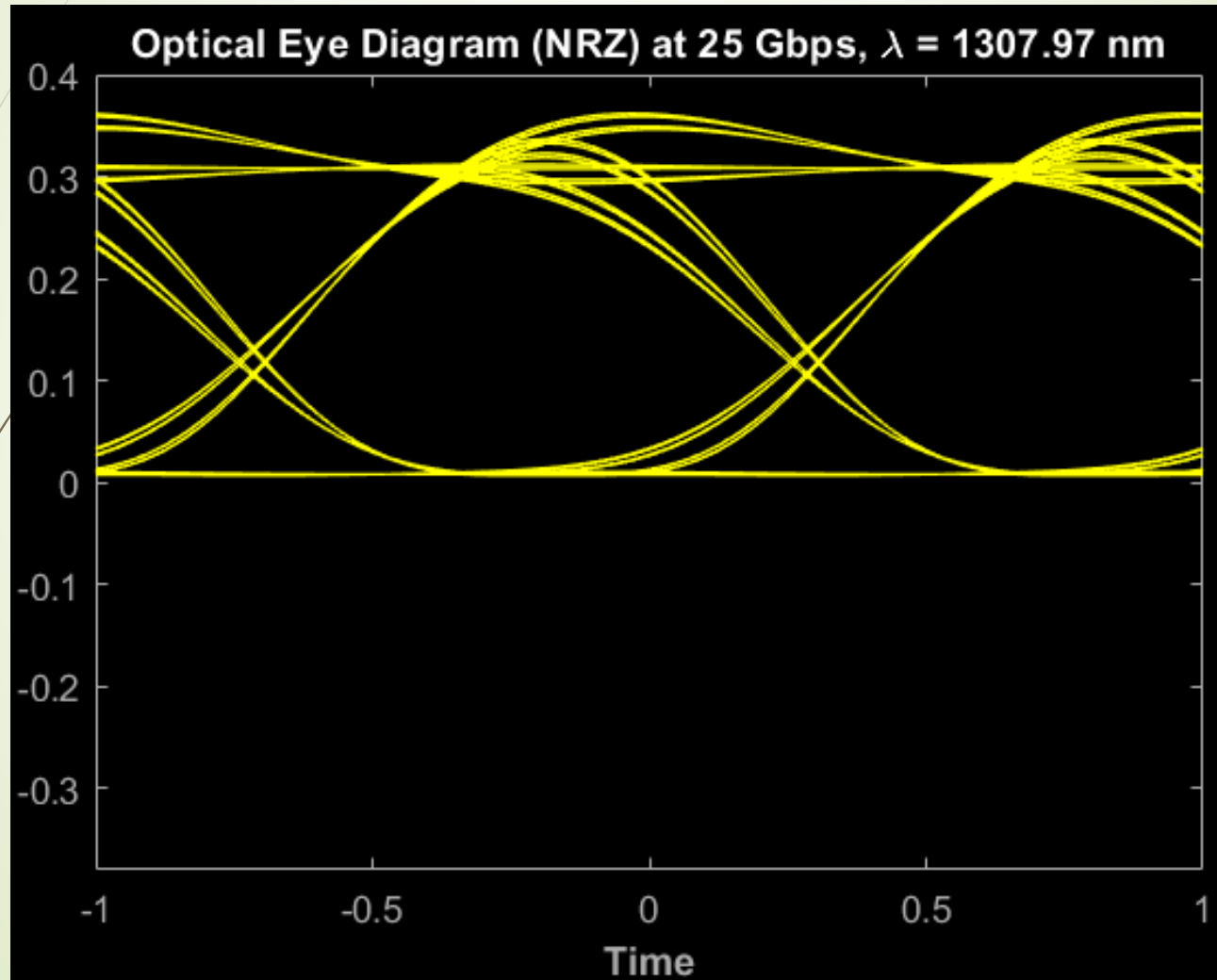
For critical coupling $r = a$

$$a^2 = \exp(-\alpha L)$$

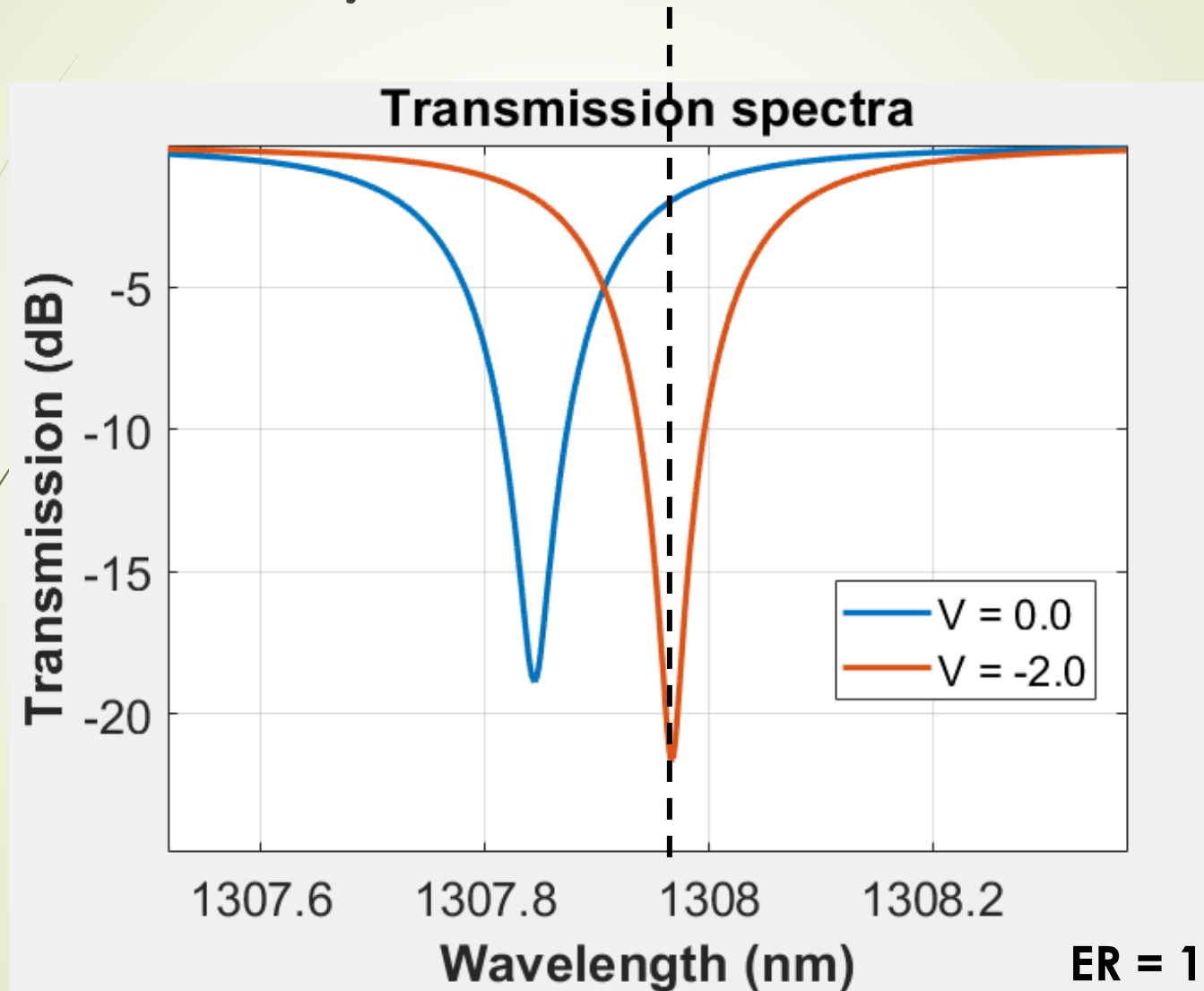
$$\text{FWHM} = \frac{(1 - ra)\lambda_{\text{res}}^2}{\pi n_g L \sqrt{ra}}$$

$$Q = \frac{\lambda_0}{\Delta\lambda_{\text{FWHM}}}$$

- Lateral PN junction used for modulation



- Lateral PN junction used for modulation



ER = 18.7 dB

IL = 2.8 dB

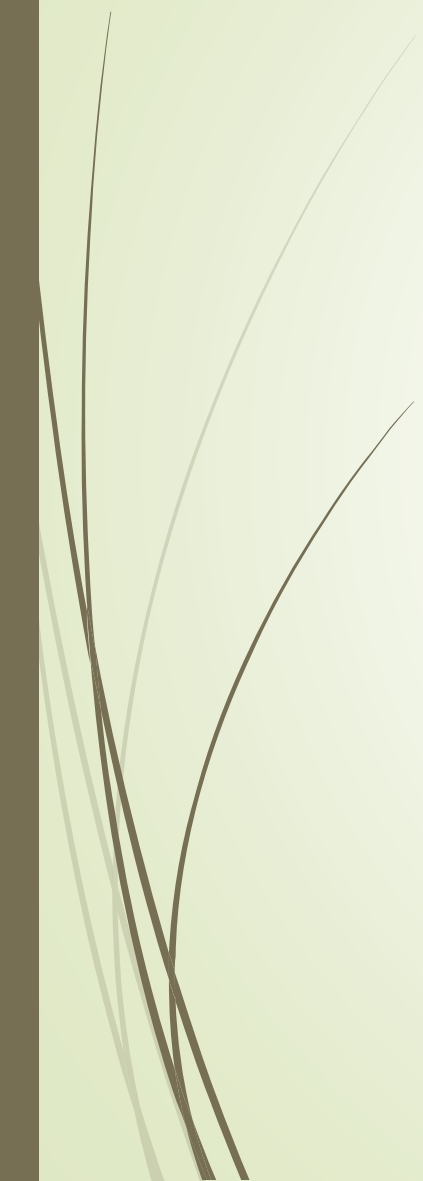


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Documentation.





Thanks 😊